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Infrared and Raman spectra of disordered materials from first principles

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Abstract

We describe a scheme for calculating the infrared and Raman spectra of large model structures from first principles. Our scheme is based on the application of a finite electric field in density functional calculations with periodic boundary conditions. Coupling tensors for infrared and Raman spectra are obtained by numerically calculating first and second derivatives of the atomic forces with respect to the electric field. The method is illustrated through a study of the vibrational properties of vitreous silica, including inelastic neutron scattering, infrared, and Raman spectroscopies.

Keywords: Infrared; Raman; Vibrational properties characterization; Density functional theory (DFT)

1. Introduction

The understanding and the prediction of the physical properties of noncrystalline solids rely on the description of their structure at the atomic level. In addition to experimental techniques which are directly sensitive to the structure, such as X-ray or neutron diffraction, detailed structural information can also be achieved from vibrational spectroscopies. For instance, Raman spectra of vitreous silica (υ-SiO_2) and of vitreous boron oxide (υ-B_2O_3) show sharp lines which have been associated to small ring structures embedded in the disordered network [1–3]. However, a detailed understanding of the vibrational spectra can only be achieved through accurate theoretical modeling.

Such high level of accuracy calls for theoretical approaches which address directly the electronic structure. In this context, first-principles calculations based on density functional theory (DFT) are particularly appealing for the good compromise between accuracy and computational cost. Indeed, model structures containing up to a few hundred atoms can nowadays be investigated. Nevertheless, first-principles simulations of infrared and Raman spectra of large model structures have remained limited, mainly because of the high computational cost for calculating coupling tensors.

A common scheme for obtaining infrared and Raman coupling tensors in bulk systems relies on treating the coupling to the electric field in a perturbational way. This scheme has been particularly successful in approaches based on plane–wave basis sets and normconserving pseudopotentials [5–10]. In particular, such a perturbational method has recently been introduced for calculating nonresonant Raman spectra of relatively large systems [11]. However, while perturbational methods carry the advantage of avoiding numerical derivatives, their practical implementation can be very involved and generally requires dedicated linear-response computer codes. For instance, the treatment of ultrasoft pseudopotentials, which give a competitive advantage in the study of large model structures, is recognized to be highly nontrivial in perturbational formulations [12,13].

In this paper, we introduce a scheme for calculating infrared and Raman spectra relying on a recently developed method which permits the application of a finite electric field in density functional calculations with periodic boundary conditions [14]. This method gives derivatives with respect to the electric field in a fully equivalent way as

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compared to ordinary perturbational methods [14,15]. However, in contrast to perturbational methods [5,11] this approach relies on standard density functional codes for the calculation of total energies, only requiring the addition of a term to the energy functional. Hence, the implementation of this method in a plane–wave scheme is relatively straightforward for both normconserving and ultrasoft pseudopotentials. We express infrared and Raman coupling tensors within this approach as finite differences of the atomic forces with respect to the electric field.

While the long term goal of such methodologies is to develop a flexible theoretical tool to address the vibrational spectra of any kind of disordered material, the description of resonant Raman spectra can at present not be achieved within this approach. This condition is particularly limiting in the study of small band-gap materials. For instance, resonant Raman spectra of amorphous carbon are known to exhibit a rich behavior for varying frequency of the exciting photon [4]. An extension of this approach to include resonant excitations remains therefore highly desirable. Nevertheless, in the case of large band-gap materials, Raman spectra are generally obtained in nonresonant conditions. In this case, the present scheme gives a complete description of the involved physical processes, since the frequency of the incoming photon is unable to excite electrons across the band gap. We therefore chose a large band-gap material such as vitreous silica (υ-SiO_2) to illustrate how this scheme can lead to a comprehensive investigation of the vibrational spectroscopies.

This paper is organized as follows. In Section 2, we briefly expose the treatment of finite electric fields in calculations with periodic boundary conditions. Section 3 gives the relevant theoretical formulae and describes our approach for calculating infrared and Raman spectra. The application to vitreous silica is found in Section 4. Conclusions are drawn in Section 5.

2. Finite electric fields

The difficulty of treating finite electric fields is related to the intrinsic nonperiodic nature of the position operator. Recent developments [14,16] within the modern theory of polarization [17,18] have demonstrated that it is possible to overcome this limitation. Several successful investigations have already been carried out with this methodology [19–21].

For a system obeying periodic boundary conditions, for which the Brillouin zone is sampled at the sole Γ point, its long-living metastable state induced by the presence of a finite electric field $\mathcal{E}$ (taken along x) can be described by the variational energy functional:

$$E^{\mathcal{E}}[\{\psi_i\}] = E^{(0)}[\{\psi_i\}] \; - \Omega\mathcal{E}\left(P^{el}[\{\psi_i\}] + P^{ion}\right), \tag{1}$$

where Ω is the volume of the unit cell and $E^{(0)}[\{\psi_i\}]$ the usual energy functional in the absence of an electric field. $P^{el}[\{\psi_i\}]$ and P^{ion} are the electronic and ionic polarizations, respectively. The electronic polarization is defined as [22]:

$$P^{el}[\{\psi_i\}] = -\frac{1}{\Omega}\frac{L}{\pi}Im(\ln det\, \mathbf{S}[\{\psi_i\}]), \tag{2}$$

where L is the periodicity of the cell and $\mathbf{S}[\{\psi_i\}]$ a matrix calculated for the set of doubly occupied wave functions $\{\psi_i\}$:

$$S_{ij} = \langle\psi_i|e^{2\pi ix/L}|\psi_j\rangle. \tag{3}$$

The ionic polarization P^{ion} is defined as:

$$P^{ion} = \frac{1}{\Omega}\sum_I Z_I X_I, \tag{4}$$

where X_I is the position coordinate of the Ith ion along the direction of the applied field and Z_I its core charge. Upon minimization of functional (1), the wave functions ψ are subject to the orthonormality constraints:

$$\langle\psi_i|\psi_j\rangle = \delta_{ij}. \tag{5}$$

Within this approach, dielectric susceptibilities, which correspond to the second derivatives of the energy with respect to the electric field, can be calculated by taking finite differences of the polarization of the system subject to electric field:

$$\chi_{ij} = \frac{\partial^2 E_{tot}}{\partial\mathcal{E}_i\partial\mathcal{E}_j} \cong \frac{\Delta P_i^{el}}{\Delta\mathcal{E}_j}, \tag{6}$$

where it is understood that the derivative with respect to the electric field is evaluated at $\mathcal{E}=0$, a convention adopted throughout this paper. The dielectric constant is then given by:

$$\epsilon_{ij}^{\infty} = \delta_{ij} + 4\pi\chi_{ij}. \tag{7}$$

For isotropic systems, we will only retain the spherical average ϵ^{∞} of this tensor.

3. Vibrational properties

For the calculation of the infrared and Raman spectra, the vibrational frequencies and their corresponding eigenmodes are derived from the dynamical matrix. Using a similar notation as Gonze and Lee [8], we express the analytical part of the dynamical matrix as:

$$D_{Ii,Jj} = \frac{1}{\sqrt{M_I M_J}}\frac{\partial^2 E_{tot}}{\partial r_{Ii}\partial r_{Jj}}, \tag{8}$$

where E_{tot} is the total energy of the system and M_I the mass of the Ith atom. Upper and lower case indices run over the atoms in the simulation cell and over the three Cartesian directions, respectively. The full dynamical matrix also contains a nonanalytical matrix accounting for vibrational excitations longitudinal to the normalized direction **q** along which the vibrational momentum is exchanged [23]:

$$\mathrm{D}^{\mathbf{q}}_{\mathrm{Ii,Jj}} = \mathrm{D}_{\mathrm{Ii,Jj}} + \mathrm{D}^{\mathrm{NA},\mathbf{q}}_{\mathrm{Ii,Jj}}, \tag{9}$$

where:

$$\mathrm{D}^{\mathrm{NA},\mathbf{q}}_{\mathrm{Ii,Jj}} = \frac{1}{\sqrt{M_\mathrm{I} M_\mathrm{J}}} \frac{4\pi}{\epsilon^\infty \Omega} \left(\sum_\mathrm{l} Z_{\mathrm{I,li}}{*} \mathrm{q_l} \right) \left(\sum_\mathrm{m} Z_{\mathrm{J,mj}}{*} \mathrm{q_m} \right). \tag{10}$$

In Eq. (10), we consider that the high-frequency dielectric tensor of our system is isotropic. The Born effective charge tensors $Z_{\mathrm{I,ij}}*$ appearing in Eq. (10) are defined as the induced polarization along the direction i by a unitary displacement of the Ith atom in the direction j [18]:

$$Z_{\mathrm{I,ij}}{*} = \Omega \frac{\partial P^{\mathrm{el}}_\mathrm{i}}{\partial r_{\mathrm{Ij}}} = \frac{\partial^2 E_{\mathrm{tot}}}{\partial \mathcal{E}_\mathrm{i} \partial r_{\mathrm{Ij}}} = -\frac{\partial F_{\mathrm{Ij}}}{\partial \mathcal{E}_\mathrm{i}}, \tag{11}$$

where the last equality gives an alternative definition of the Born effective charge tensors in terms of the atomic forces $\mathbf{F}_\mathrm{I}$.

For a given choice of **q**, the frequencies ω_n and the corresponding normalized eigenmodes $\upsilon^\mathrm{n}_{\mathrm{Ii}}$ are obtained by diagonalizing the dynamical matrix. The atomic displacements $u^\mathrm{n}_{\mathrm{Ii}}$ associated with the mode n, are then given by:

$$u^\mathrm{n}_{\mathrm{Ii}} = \frac{1}{\sqrt{M_\mathrm{I}}} \upsilon^\mathrm{n}_{\mathrm{Ii}}. \tag{12}$$

In this approach, we obtain the analytical part of the dynamical matrix by taking finite differences of atomic forces [24,25]. Furthermore, using the possibility of applying a finite electric field [14], we obtain the Born effective charge tensors $Z_{\mathrm{I,ij}}*$ by taking finite differences of atomic forces $\mathbf{F}_\mathrm{I}$ with respect to the electric field:

$$Z_{\mathrm{I,ij}}{*} = -\frac{\Delta F_{\mathrm{Ij}}}{\Delta \mathcal{E}_\mathrm{i}}. \tag{13}$$

3.1. Infrared spectra

We now address the real and imaginary parts of the dielectric response function in the long wavelength limit [26]. Since these dielectric properties involve transverse vibrational excitations, we use vibrational frequencies and eigenmodes pertaining to the analytical part of the dynamical matrix. The real part of the dielectric response function $\epsilon_1(\omega)$ can then be expressed as:

$$\epsilon_1(\omega) = \epsilon^\infty + \frac{4\pi}{3\Omega} \sum_\mathrm{n} \frac{|\boldsymbol{f}^\mathrm{n}|^2}{\omega^2 - \omega_\mathrm{n}^2}, \tag{14}$$

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where the oscillator strengths $\boldsymbol{f}^\mathrm{n}$ are given by:

$$f^\mathrm{n}_\mathrm{k} = \sum_{\mathrm{I,i}} Z_{\mathrm{I,ki}}{*} u^\mathrm{n}_{\mathrm{I,i}}. \tag{15}$$

The imaginary part of the dielectric constant $\epsilon_2(\omega)$ is given by [26]:

$$\epsilon_2(\omega) = \frac{4\pi^2}{3\Omega} \sum_\mathrm{n} \frac{|\boldsymbol{f}^\mathrm{n}|^2}{2\omega_\mathrm{n}} \delta(\omega - \omega_\mathrm{n}). \tag{16}$$

The dielectric function described above gives access to all the dielectric properties. In particular, the energy loss function is obtained as $-Im[1/\epsilon(\omega)]$. However, we found it convenient to access the latter function by a direct calculation:

$$-Im[1/\epsilon(\omega)] = \frac{4\pi^2}{\Omega(\epsilon^\infty)^2} \sum_\mathrm{n} \frac{(\mathbf{q}\cdot\boldsymbol{f}^\mathrm{n})^2}{2\omega_\mathrm{n}} \delta(\omega - \omega_\mathrm{n}), \tag{17}$$

where all the quantities are calculated for frequencies ω_n and vibrational displacements $u^\mathrm{n}_{\mathrm{Ii}}$ obtained by diagonalizing the full dynamical matrix $\mathrm{D}^{\mathbf{q}}_{\mathrm{Ii,Jj}}$ given in Eq. (9). For isotropic systems, the energy loss function can be averaged over all directions of **q**. Here, we use the three Cartesian directions for this average.

3.2. Raman spectra

In a Raman scattering process, an incoming photon of frequency ω_L and polarization $\boldsymbol{e}_\mathrm{L}$ is scattered to an outgoing photon of frequency ω_S and polarization $\boldsymbol{e}_\mathrm{S}$ either creating (Stokes process) or annihilating (anti-Stokes process) a phonon of frequency ω_n. We here consider nonresonant Raman scattering processes which can be described in the Placzek approximation. The reduced Raman cross section reads [26,27]:

$$\frac{\mathrm{d}\sigma(\omega)}{\mathrm{d}\Omega} \sim \sum_\mathrm{n} |\boldsymbol{e}_\mathrm{S} \cdot \boldsymbol{\alpha}^\mathrm{n} \cdot \boldsymbol{e}_\mathrm{L}|^2 \delta(\omega - \omega_\mathrm{n}), \tag{18}$$

where the Raman tensors $\boldsymbol{\alpha}^\mathrm{n}$ are defined as [27]:

$$\alpha^\mathrm{n}_{\mathrm{ij}} = \sqrt{\Omega} \sum_{\mathrm{Ik}} \frac{\partial \chi_{\mathrm{ij}}}{\partial r_{\mathrm{Ik}}} u^\mathrm{n}_{\mathrm{Ik}}, \tag{19}$$

where χ_{ij} is the dielectric susceptibility tensor. The third rank tensors $\partial\chi_{\mathrm{ij}}/\partial r_{\mathrm{Ik}}$ can be expressed in terms of second derivatives of the atomic forces with respect to the electric field:

$$\frac{\partial \chi_{\mathrm{ij}}}{\partial r_{\mathrm{Ik}}} = \frac{\partial^2 P^{\mathrm{el}}_\mathrm{i}}{\partial r_{\mathrm{Ik}} \partial \mathcal{E}_\mathrm{j}} = \frac{1}{\Omega} \frac{\partial^3 E_{\mathrm{tot}}}{\partial r_{\mathrm{Ik}} \partial \mathcal{E}_\mathrm{i} \partial \mathcal{E}_\mathrm{j}} = -\frac{1}{\Omega} \frac{\partial^2 F_{\mathrm{Ik}}}{\partial \mathcal{E}_\mathrm{i} \partial \mathcal{E}_\mathrm{j}}. \tag{20}$$

We obtain the third rank tensors $\partial\chi_{\mathrm{ij}}/\partial r_{\mathrm{Ik}}$ by applying finite electric fields [14] and calculating the derivatives of

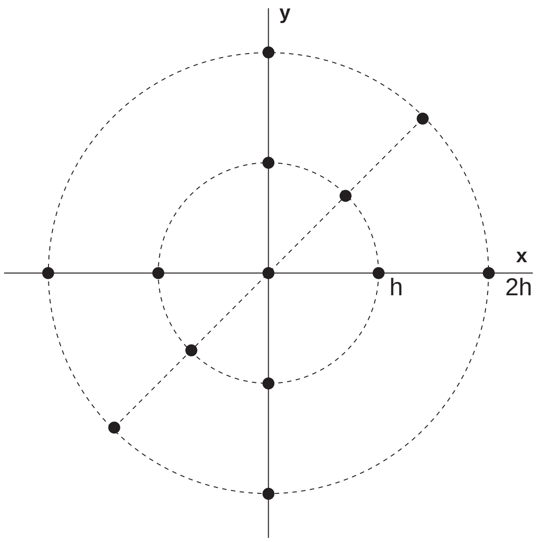

Fig. 1. Schematic diagram illustrating the electric fields in the *xy* plane which were used for the numerical derivatives. Electric fields in the *yz* and *zx* lanes were chosen similarly. Hence, all the results in this work could be obtained through just 25 different electronic minimizations in presence of an electric field.

the atomic forces numerically. We obtain the diagonal terms $\partial\chi_{ii}/\partial r_{Ik}$, using a 5-point formula for second-order numerical derivatives. Dropping the indices of the atomic forces, this formula reads:

$$\frac{d^2F}{d\mathcal{E}_i^2}(0) = \frac{1}{12h^2}[-F(-2h) + 16F(-h) - 30F(0) + 16F(h) - F(2h)], \quad (21)$$

where the forces are calculated for electric fields along the Cartesian direction i with intensities $\mathcal{E}_i = 0, \pm\mathrm{h}, \pm 2\mathrm{h}$. For deriving the off-diagonal terms, we apply electric fields with equal components along two Cartesian axes (see Fig. 1). We then derive the mixed derivatives by taking advantage of the Taylor expansion of the atomic forces with respect to two electric fields $\mathcal{E}_i$ and $\mathcal{E}_j$:

$$F(\mathcal{E}_i,\mathcal{E}_j) = F(0,0) + \frac{\partial F}{\partial\mathcal{E}_i}\mathcal{E}_i + \frac{\partial F}{\partial\mathcal{E}_j}\mathcal{E}_j + \frac{1}{2}\frac{\partial^2F}{\partial\mathcal{E}_i^2}\mathcal{E}_i^2 + \frac{1}{2}\frac{\partial^2F}{\partial\mathcal{E}_j^2}\mathcal{E}_j^2 + \frac{\partial^2F}{\partial\mathcal{E}_j\partial\mathcal{E}_j}\mathcal{E}_i\mathcal{E}_j + \ldots \quad (22)$$

By taking $\mathcal{E}_i = \mathcal{E}_j = \lambda$, the mixed term $\partial^2F/\partial\mathcal{E}_i\partial\mathcal{E}_j$ is then given by:

$$\frac{\partial^2F}{\partial\mathcal{E}_i\partial\mathcal{E}_j} = \frac{1}{2}\frac{\partial^2F}{\partial\lambda^2} - \frac{1}{2}\frac{\partial^2F}{\partial\mathcal{E}_i^2} - \frac{1}{2}\frac{\partial^2F}{\partial\mathcal{E}_j^2}. \quad (23)$$

In this way, we obtain all the tensors $\partial\chi_{ij}/\partial r_{Ik}$ by performing 25 selfconsistent minimizations of the electric-field dependent energy functional. Since a minimization provides the atomic forces for all the atoms at once, the required number of such minimizations does not depend on the number of atoms.

4. Application to vitreous silica

The vibrational spectroscopies of vitreous silica have already been the focus of several first-principles studies [2,12,24,25,28]. In particular, infrared and Raman spectra were calculated through concepts pertaining to the Berry phase theory of polarization [28] and through a perturbational approach [12,13], respectively. We here illustrate the potential of our scheme based on finite electric fields, which bears the important computational advantage of being significantly less costly.

While the previous studies used a model structure of vitreous silica generated by ab initio molecular dynamics [29,30], we here focus on a different model structure, first generated by classical molecular dynamics [31] and subsequently fully relaxed within a first-principle scheme [12]. The model structure is periodic and is defined by a cubic simulation cell containing 72 independent atoms. The size of the simulation cell was set to match the experimental density (2.2 g/cm^3). The bonding structure consists of a regular network of corner-sharing tetrahedra, the Si atoms being at their center and the O atoms at their corners. The structure contains a single four-membered ring and no three-membered rings, to comply with the low experimental concentrations of such rings [12]. The model is characterized by an average Si–O–Si bond angle of 152°, close to the experimental estimate of 151° [32]. The comparison between the calculated and measured [33] total neutron structure factor in Fig. 2 supports the validity of this model structure. The theoretical structure factor in Fig. 2 was calculated using Debye–Waller factors derived from the vibrational properties in the harmonic approximation [34].

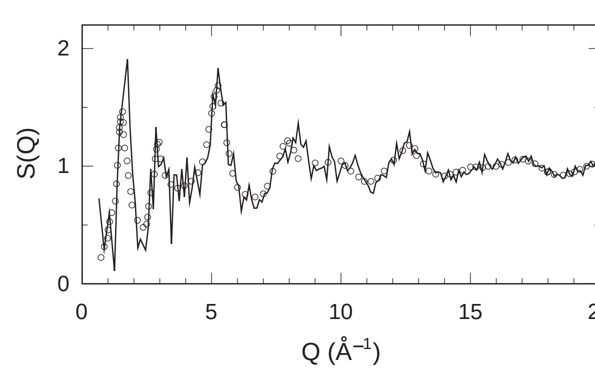

Fig. 2. Total neutron structure factor of our model structure of vitreous silica (solid) compared to experiment (circles) [33]. The calculated structure factor is obtained for a temperature of 300 K in the harmonic approximation [34].

All the calculations in this paper are carried out within a density functional scheme based on plane–wave basis sets and ultrasoft pseudopotentials [35,36]. Settings for the calculations of the electronic structure and the vibrational properties are the same as in previous calculations [12,28]. In the finite-field calculations, we varied the electric field by strides of h=1.10^{-2} a.u., remaining within the linear regime.

Before addressing the infrared and Raman spectra, we validate the vibrational properties of our model structure investigating the effective neutron density of states, since this spectrum most closely resembles the actual vibrational density of states. We calculated the effective neutron density of states following the scheme in Refs. [24,25]. In Fig. 4, the actual vibrational density of states and the effective neutron density of states are compared to experimental results [37,38], showing overall good agreement. The agreement between theory and experiment is similar to that found previously [24,25] for a model structure generated by first-principles molecular dynamics [29,30].

We calculated the dielectric response functions using finite electric fields as described in the previous sections. The high-frequency dielectric constant ϵ^{∞} of our model was found to be 2.0, in good agreement with the experimental value (2.1, from Ref. [39]). We calculated $\Delta\epsilon=\epsilon^0-\epsilon^{\infty}=1.76$ [28], resulting in a static dielectric constant ϵ^0=3.8, in excellent agreement with the experimental value (3.8, from Ref. [40]). In Fig. 4, the calculated response functions are compared with the experimental spectra [39]. The real and imaginary parts of the dielectric function show two principal resonances at ~440 and ~1090 cm^{-1}, in good agreement with the experimental resonances at 460 and 1075 cm^{-1} [39]. The third resonance observed experimentally at 800 cm^{-1} corresponds to the feature at ~750 cm^{-1}, as can be inferred by comparison with the effective neutron density of states where a corresponding peak appears. The theoretical curves also show spurious features at ~860 and ~1000 cm^{-1}. These features originate from distorted $Si(O_{1/2})_4$ tetrahedra with O–Si–O bond angles larger than ~120° and are due to structural strain in our model structure. We report in Table 1, the peak positions of the three main resonances. The TO and LO frequencies are derived from the imaginary part of the dielectric function and from the energy loss function $-Im[1/\epsilon(\omega)]$, respectively. We found LO–TO splittings of 80, 8 and 194 cm^{-1} for resonances of increasing frequency, in fair agreement with the respective experimental splittings of 40, 20, and 175 cm^{-1}.

Using Raman coupling tensors calculated with the scheme based on finite electric fields, we obtained reduced

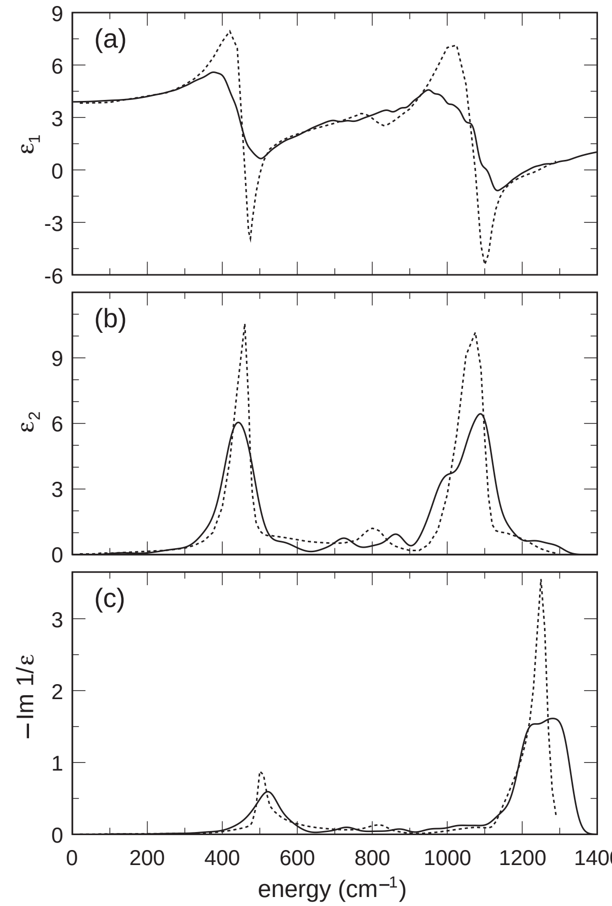

Fig. 4. Calculated dielectric response functions (solid) for our model of vitreous silica: (a) real and (b) imaginary part of the dielectric function, and (c) energy loss function. The calculated results are compared with the experimental data of Ref. [39] (dotted). The theoretical curve in (a) has been broadened by a Lorentzian function with Γ=25.6 cm^{-1}. A Gaussian broadening (22 cm^{-1}) is used for the theoretical curves in (b) and (c).

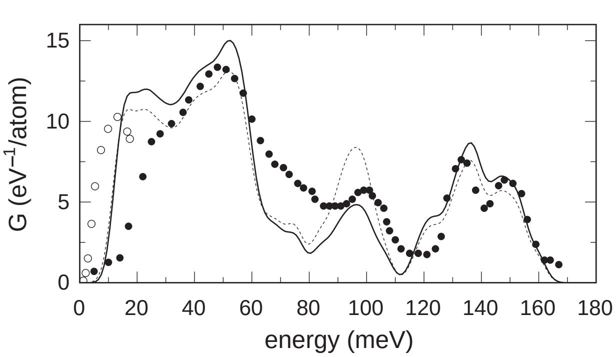

Fig. 3. Calculated effective neutron density of states (solid) for our model structure of vitreous silica, compared to experimental results from Refs. [37] (closed symbols) and [38] (open symbols). The actual vibrational density of states (dotted) is also provided for comparison. A Gaussian broadening (2 meV) is used for the theoretical spectra.

Table 1
Frequencies (in cm^{-1}) of TO and LO peaks as deduced from the imaginary part of the dielectric function (TO) and the energy loss function (LO)

Theory		Experiment	
TO	LO	TO	LO
442	522	460	500
724	732	800	820
1088	1282	1075	1250

Experimental data are from Ref. [39].

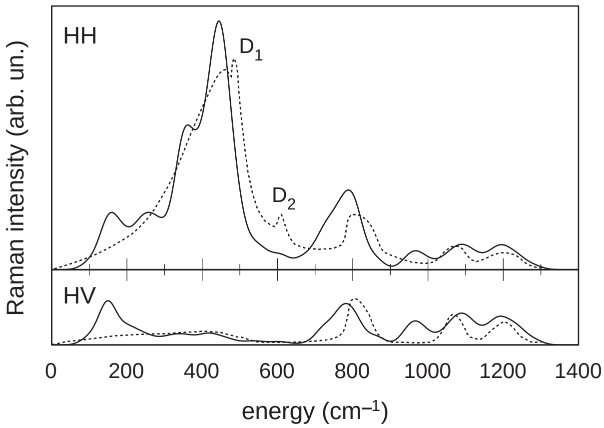

Fig. 5. Calculated reduced HH and HV Raman spectra (solid) for our model of vitreous silica, compared with the experimental spectra of Ref. [1] (dotted). The calculated spectra are scaled to match the integrated intensity of the experimental HH spectrum. For clarity, the curves in the lower panel are magnified by a factor of two. A Gaussian broadening (22 cm^{-1}) is used for the theoretical spectra.

Raman spectra for our model structure of vitreous silica. We considered the cases of parallel (HH) and orthogonal (HV) polarizations of in- and out-going photons. Our calculated spectra are given in Fig. 5, where they are compared with corresponding experimental results [1]. The main peaks observed in the experimental HH spectrum, i.e. the large peak at ~450 cm^{-1}, the peak at 800 cm^{-1}, and the high-frequency doublet (~1075 and ~1200 cm^{-1}), are overall well reproduced by our calculation. The D_2 line corresponds to an in-phase breathing motion of oxygen atoms in three-membered rings [2]. The calculated spectrum does not reproduce this feature since our structural model does not contain any three-membered ring [2]. The D_1 line corresponds similarly to four-membered rings [2]. While a single four-membered ring is present in our model structure, the width of the corresponding line is too large for this line to stand out in the calculated spectrum [12]. The calculated spectrum also shows a peak at ~150 cm^{-1} which does not find a counterpart in the experimental spectrum. This peak corresponds to the lowest peak in the vibrational density of states (Fig. 3), where it occurs at larger energies than for the experimental data of Ref. [38]. We attribute this shift towards larger frequencies to structural strain in our model. It is interesting to note that, for the model structure of Refs. [29,30], a similar peak occurred at lower energies in the vibrational density of states [24,25], giving rise to only a weak feature in the corresponding Raman spectrum [12]. The peak at ~1000 cm^{-1} results from distorted tetrahedra, as discussed above. Overall, the same comments also apply for the comparison between calculated and measured HV spectra. Note, in particular, that the intensities in Fig. 5 have been set by comparison with the experimental HH spectrum. Hence, the good agreement for the overall intensity of the HV spectrum implies that the relative intensity between HH and HV spectra is well reproduced in our calculation.

5. Conclusion

We introduced a scheme for calculating vibrational spectra of disordered materials from first principles. The scheme relies on the application of finite electric fields in density functional calculations with periodic boundary conditions [14]. Through numerical derivatives with respect to the field, all the relevant coupling tensors determining infrared and Raman spectra are evaluated with a small computational overhead with respect to the calculation of the vibrational eigenmodes. We illustrated the scheme through an application to a model structure of vitreous silica. By comparing theory and experiment for three different vibrational spectroscopies (inelastic neutron scattering, infrared, and Raman), we handle a powerful computational tool for the interpretation of vibrational spectra, and, consequently, for the understanding of the underlying structure. We are convinced that the generalized application of the present scheme will offer great potential for unveiling the short- and intermediate-range structure of a large variety of disordered materials, including disordered forms of carbon and silicon carbide.

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